

A Comparison of Gesture-based Interaction and Controller-based Interaction for External Users in Co-located Asymmetric Virtual Reality

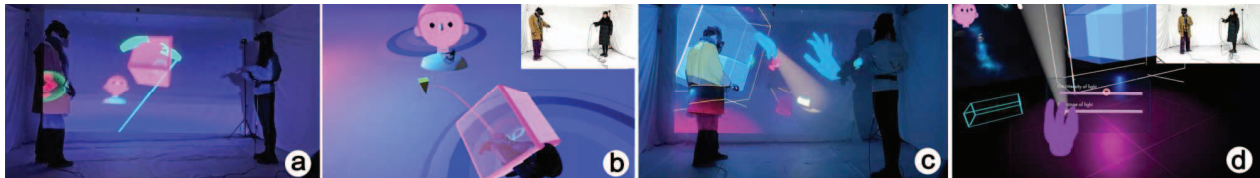
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Figure 1: The Gesture-based interaction (a)(c) and the controller-based interaction (b)(d) in scenario "Object Transition", "Collaborative Game" respectively.

ABSTRACT

Head-mounted displays (HMDs) create a highly immersive experience, while limit VR users' awareness of external users that co-located in the same environment. In our work, we designed a wearable gesture-based interface for external users in two main asymmetric VR scenarios: (1) Object Transition, (2) Collaborative Game. We conducted a hybrid user study with twenty participants to compare the gesture-based interaction and controller-based interaction to investigate the effects on VR experience, social presence, and collaborative efficiency for external users. Our study revealed that the participants in gesture-based interaction received more positive feedback compared to those in controller-based interaction.

Keywords: Sharing VR experience, Asymmetric interaction, Gesture-Based interaction.

Index Terms: Index Terms: Human-centered computing—Human computer interaction (HCI) — Interaction paradigms — Virtual reality

1 INTRODUCTION

Head-mounted displays provide the user with a highly immersive VR experience. However, the HMD separates VR users from the physical world, causing communication gaps with external users who co-locate at the same environment. Researchers present various interfaces that expand the virtual world into physical environment to generate asymmetric VR experience. Previous research on asymmetric VR experience mainly collaborates with VR users by using controllers and portable screen [1, 2]. However, these methods provide less immersion and limit the external users' interaction. In our study, we aimed to design a novel system that overcomes these shortages to enhance the external users' sense of presence and their freedom of interaction. We propose a co-located VR experience in two main scenarios, (1) Object Transition and (2) Collaborative Game, in

which we compared the gesture-based interaction and the controller-based interaction for external users.

For the gesture-based interaction, the external user uses a VIVE tracker that attached to his or her head to locate the relative position in the virtual scenario. An accompanying virtual camera projects the external user's perspective onto the physical environment. We use Leap Motion that tied around the external user's waist to interact with the VR user in real-time. For the controller-based interaction, the external user uses a VIVE hand controller that attached a 7-inch removable screen as a virtual camera to perform the tasks. In a user study ($N = 20$), we compared the participants in both conditions. The gesture-base interaction showed significant advantage on VR experience, social presence, and collaborative efficiency.

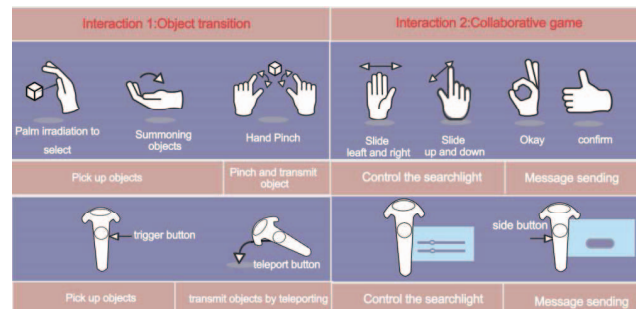


Figure 2: The instructions of gesture-based interaction and controller-based interaction in two scenarios.

2 OBJECT TRANSITION

The selection and transition of virtual objects are important interactive behaviours in VR collaboration [3]. Therefore, we designed Object Transition. In this task, both VR and external users should pick up specific colour notes and throw them to each other. We provided object interaction through generating gesture interactive rays (Figure 1.a.b) in gesture-based interaction, which includes three hand gestures: click, summon, and pinch (Figure 2). The external user can select the blue note through a generated blue ray by pinching with five fingers, then he or she can summon the note ball by flexing the four fingers and placed into the square to throw at the VR user by pinching hand as feedback. The VR user can place the purple note into the square and pass it back to the external

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user. For the controller-based interaction, it also includes the interaction of picking up and transmitting the notes. The external user can press the handle button to pick the blue notes into the box and use the trigger button to teleport himself nearby the VR user.

3 THE COLLABORATIVE GAME

The collaborative game is a typical interactive form in asymmetric VR [4]. Collaboration can be facilitated by empowering external users and VR users with different capabilities. Therefore, we designed an asymmetric cooperative game. There is a difference between using gesture-based interaction and controller-based interaction to send messages and controlling a searchlight (Figure 1.c.d). For gesture-based interaction, the external user looks for the location of the puzzles and send gestures to inform the VR user where the puzzles are (Figure 2). The VR user can teleport and pick up the puzzles in the virtual scenario only when the external user lights up the puzzles. We designed gestures: *slide*, *okay* and *confirm*. The external user controls the range and brightness of the searchlight by pinching the index finger and thumb from the right hand and send an *Okay gesture* to inform the VR user that the puzzle has been found. In the controller-based interaction, the external user can slide the slider to control the searchlight and click the UI buttons to send messages by using the controller.

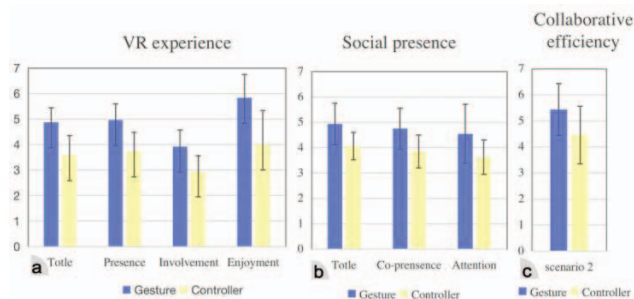


Figure 3: Results of VR experience, Social presence and Collaborative efficiency.

4 PRIMARY STUDY

We conducted a primary study with 20 participants to study the VR experience, social presence and collaborative efficiency of the gesture-based interaction and the controller-based interaction. The study was carried out in a 30-square-meter indoor space. The participants were 9 males and 11 females with the average age of 22.10 (SD = 2.3). The participants were divided into ten groups (in pairs) and were randomly assigned as a VR user or an external user. All groups tested both gesture-based interaction and controller-based interaction in two virtual scenarios. The users need to complete the collaborative efficiency after completing each virtual scene. Then they need to fill the overall experience questionnaire for each group after completing a set of tests, including VR experience, social presence (NM-Mosp). Last, they were invited in a semi-structured interview. In order to prevent the order of experiments from affecting the user experience, we will change the order of experiments after each set of tests.

4.1 Results

We evaluated the VR experience, social presence and collaborative efficiency for each system by using the 7-point Likert scale questionnaire. we conducted the analysis by the Paired Samples t-test. The results showed that the VR experience for the participants in the gesture-based interaction was significantly higher than those in the controller-based interaction

(Figure 3.a) ($t(20) = 3.913, p = 0.004$). The gesture-based interaction provided external user with higher presence, involvement and enjoyment.

For the social presence, the gesture-based interaction also reported significantly higher interaction experience in comparison with that of the controller-based interaction (Figure 3.b) ($t(20) = 2.790, p = 0.021$). The gesture-based interaction has an advantage over the controller-based interaction with regard to co-presence experience and attentional allocation. However, the gesture-based interaction did not show an advantage in terms of behavioral dependence and behavioral understanding. According to the interviews, we find that the gesture-based interaction can provide users with a stronger sense of coexistence and immersion, thereby helping them immerse themselves in the VR experience better. However, the feedback from the controller-based interaction is more direct and clearer. Therefore, the external users can understand each other's interactive behavior easily.

For collaborative efficiency, in "Collaborative Game", the gesture-based interaction reported significantly higher scores in comparison with the controller-based interaction (Figure 3.c) ($t(20) = 2.516, p = 0.033$). The external users use natural hand gestures to transmit information and cooperate more efficiently than the controller-based interaction. Due to the learning cost and accuracy rate of gesture, the task coordination efficiency of gesture-based interaction in "Object transition" mode isn't significant.

5 CONCLUSION

In this paper, we conducted a hybrid user study in the two scenarios by comparing the gesture-based interaction and the controller-based interaction in asymmetric VR. A primary study showed that our system received positive feedback on the sense of VR experience, social presence, and collaborative efficiency. The external user had a stronger sense of immersion and social experience due to the freedom of gesture and body perception. It could help both users perceive each other's interactions more clearly and give better feedback for collaboration. Meanwhile, our system still needs to be further improved in terms of learning cost and gesture design accuracy. The participants provided additional suggestions for future implementation: 1) compare gestures-based interaction and controller-based interaction in different collaborative scenarios, 2) designing more natural and cognitive gesture-based interaction.

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